

Table 2: A few popular decomposition algorithms

H / A / Alpha	Huynen	Barnes	Cloude	Holm	An & Yang	Bhattacharya & Ferry
Freeman	Neumann	Arri	Van Zyl	Singh 4	Yamaguchi	L. Zhang
Touzi	RVoG	Raney	Entropy	Anisotropy	EigenValues	Aghababae

Table 3: A few segmentation algorithms

H / A / Alpha classification	H / u / v classification	H / A / Alpha - Wishart classification	Unified Huynen classification
G.P.F. Supervised classification	Fuzzy - H / Alpha classification	Classification Rule-based Hierarchical classification	SVM Supervised classification

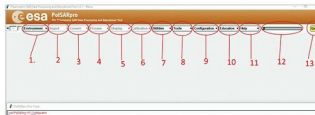


Figure 3: PolSARpro home screen

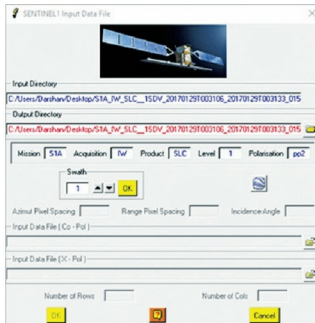


Figure 4: Sentinel window

The first step is to select the environment. The first active tab shown in Figure 3 gives options to select a single data set, dual data set or multi-data set. Select the single data set if you want to process only a single scene or a single SAR image. Select an appropriate folder location. The environment folder selection depends on a particular satellite that one wants to work with. Let's take the example of Sentinel-1 SAR SLC level 1 data (Sentinel data is freely available as discussed earlier). After selecting the environment, the import tab (No. 2 in Figure 3) will be activated. For Sentinel-1, select *Import* → *Spaceborne Sensors* → *Sentinel-1* → *Unzipped data product*.

Select the swath number, as Sentinel data contains three swaths in a single data set. This window differs according to



Figure 5: Showing C2 matrix



Figure 6: KML file (This snapshot is taken from Google Earth software developed by Google Inc.)

different sensors. It will take time to process depending on the user's hardware and the size of the data. After importing data, select *Import* → *Extract PolSAR images*. This will give you the option to create a covariance matrix. If it is quad pole data, then multiple options like T3, C3, C2, Sinclair, etc., will be shown. These matrices are the 'BASE' for further processing of the same data, and are now keys for further processing like decomposition, classification, etc. All the tabs will now be activated, including 'Process' (No. 4 in Figure 1). Make sure that the matrix you are working on appears at the bottom left, as shown in Figure 5. Whatever process one selects will be applied to the matrix mentioned in the box (C2, in this example).

When you press the 'Process' tab, all the possible speckle filtering algorithms, decomposition algorithms, classification algorithms, etc., will be visible. Now, from here it's up to you! You can try as many decompositions, filters and classifications that interest you or are relevant to your field.

One of the features of PolSARpro is that it creates a KML file which can be seen on the Google Earth application. The Google Earth app can also be linked with PolSARpro software during installation. This KML file shows the highlighted location of the SAR image on Google Earth. A sample of the highlighted image is shown in Figure 6.

A few segmentation algorithms are listed in Table 3.

Scatter plots can also be generated using any two entities such as the X-axis and Y-axis.

In short, there is a lot more that you can do with PolSARpro, which is not possible to cover in this article. The more you explore, the more functionalities you will discover. Have a happy time exploring PolSARpro! 

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